What is NumPy ?

- NumPy Stands For Numerical Python.
- NumPy is a Python library used for working with array.
- NumPy is the fundamental package for scientific computing in Python.
- It also has functions for working in domain of linear algebra, Fourier transform, and matrices.
- NumPy was created in 2005 by Travis Oliphant. It is an open-source project, and you can use it freely.

Why Use NumPy ?

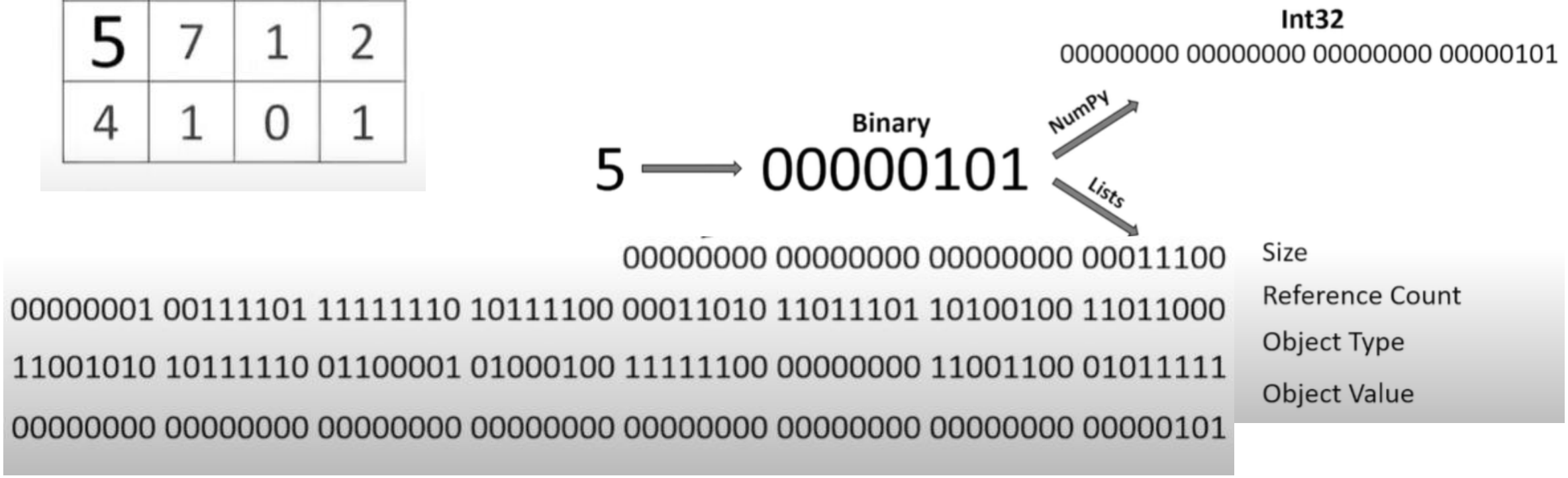
- In Python we have lists that serve the purpose of arrays, but they are slow to process.
- NumPy aims to provide an array object that is up to 50x faster than traditional Python lists.
- The Array object in NumPy is called ndarray, it provides a lot of supporting functions that make working with ndarray very easy.
- Arrays are very frequently used in data science, where speed and resources are very important.

Why is NumPy Faster Than Lists?

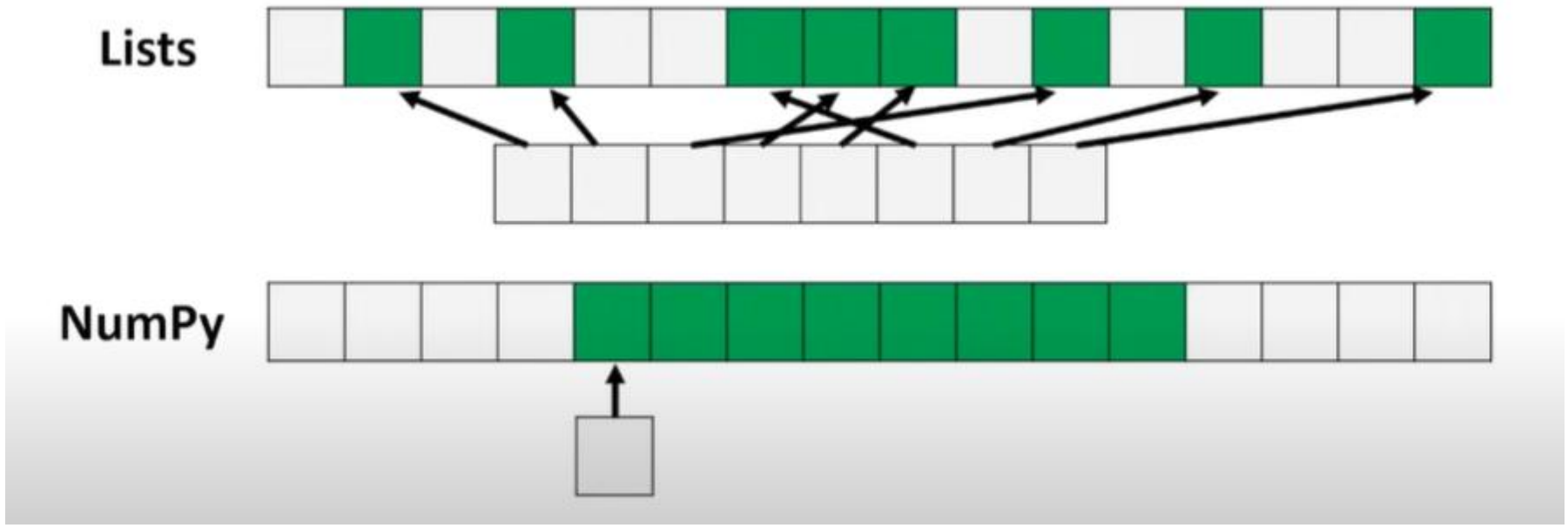
- NumPy array elements have fixed types.
- NumPy arrays are stored at one continuous place in memory unlike lists, so processes can access and manipulate them very efficiently.
  - This behavior is called locality of reference in computer science.

Why is NumPy Faster Than Lists? (Fixed Type)

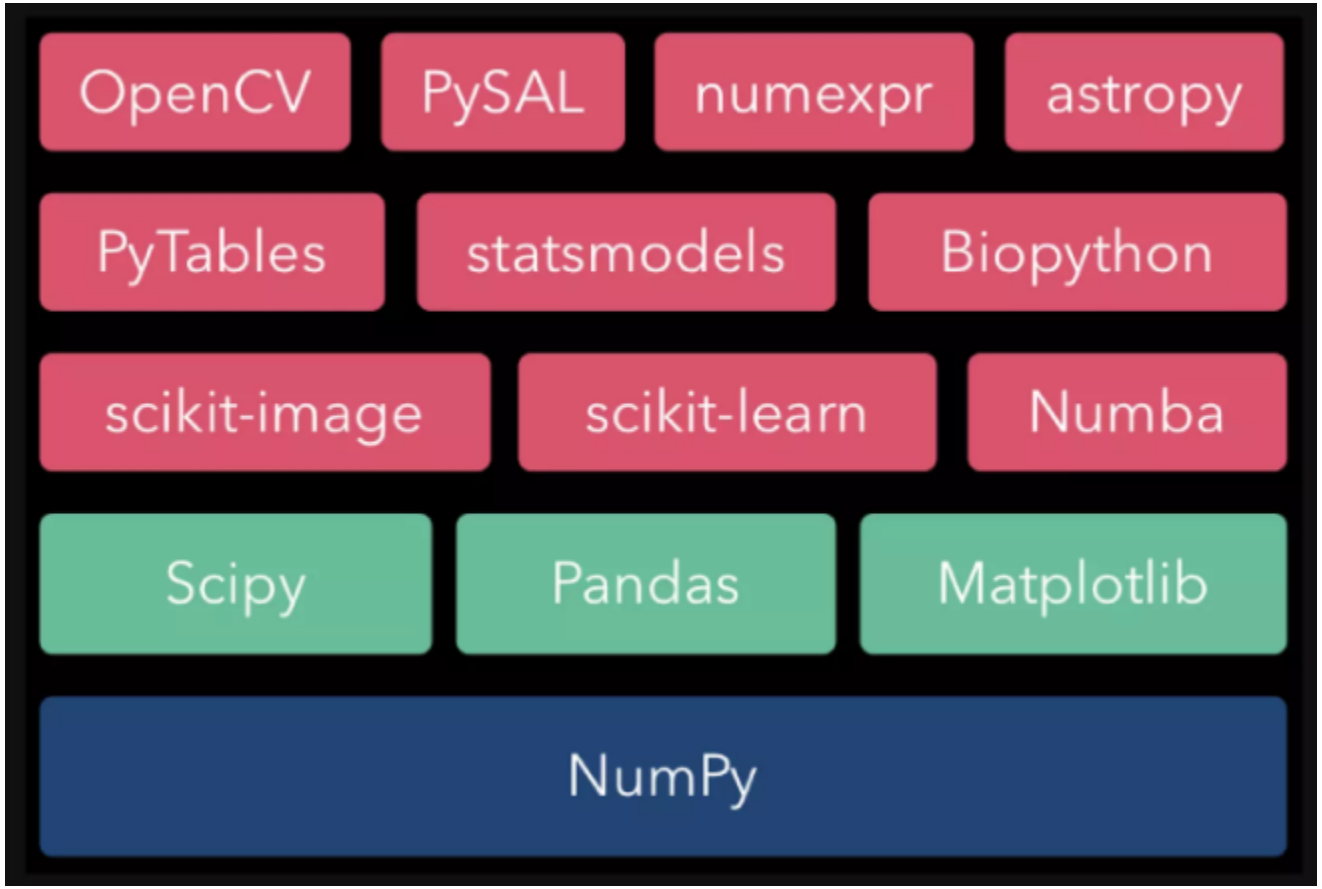
3	1	2	4
5	7	1	2
4	1	0	1



Why is NumPy Faster Than Lists? (Contiguous Memory)



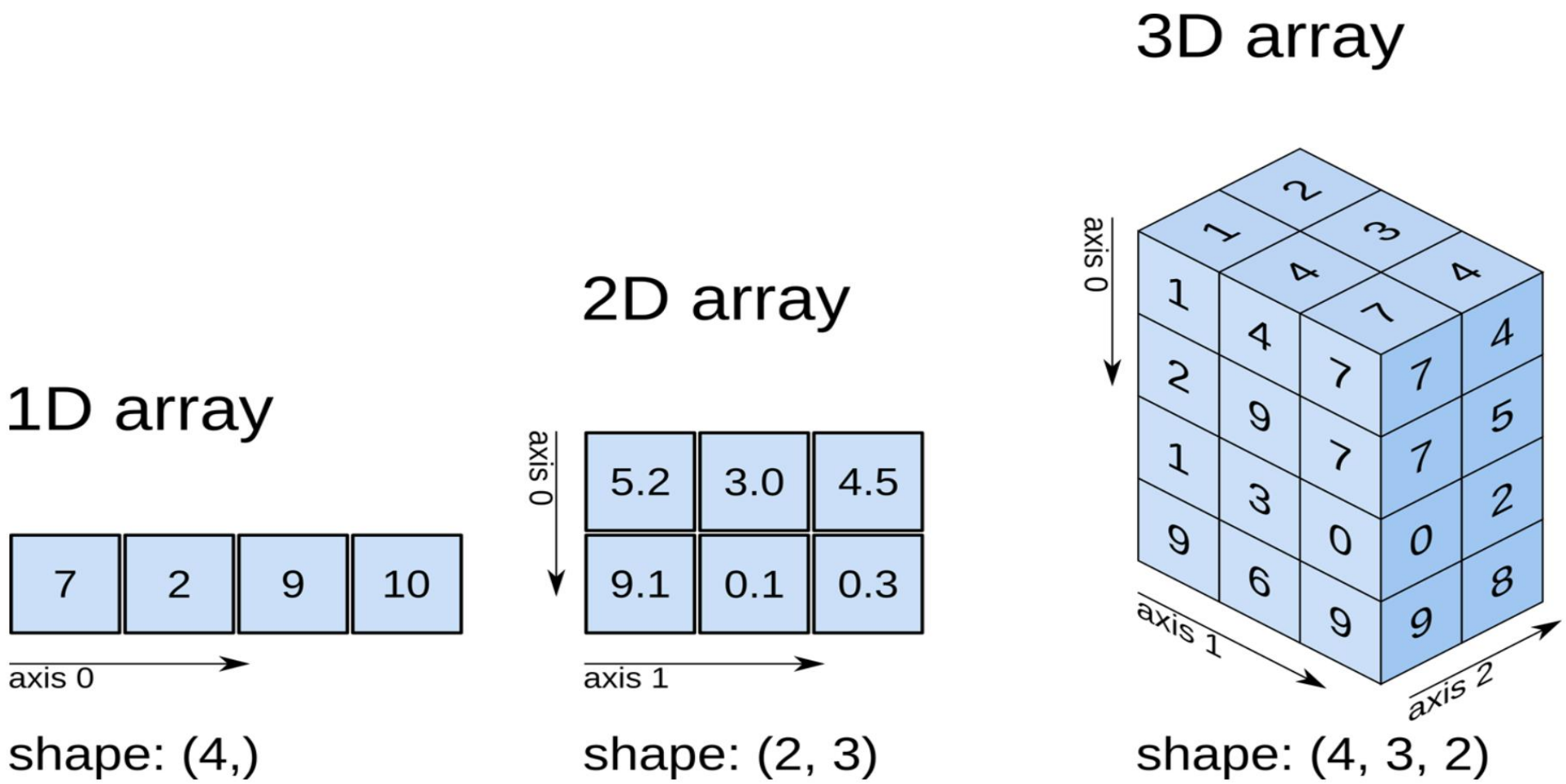
NumPy Ecosystem



Arrays in NumPy

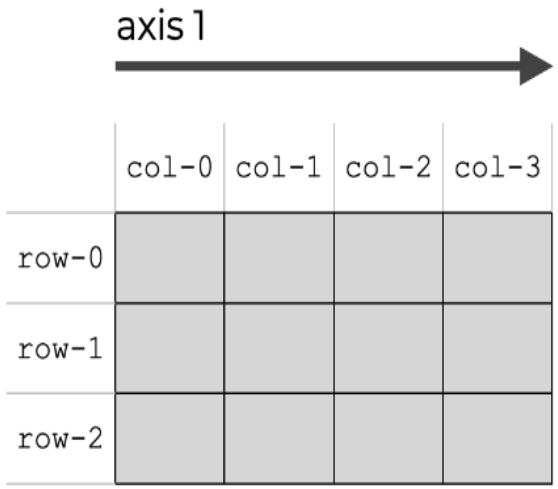
- An Array is a collection of values, organized in a specific order.
- NumPy's main object is the homogeneous multidimensional array.
- It is a table of elements (usually numbers), all of the same type, indexed by a tuple of positive integers.
- In NumPy dimensions are called axis. The number of axis is rank.
- NumPy's array class is called ndarray. It is also known by the alias array.

Arrays in NumPy

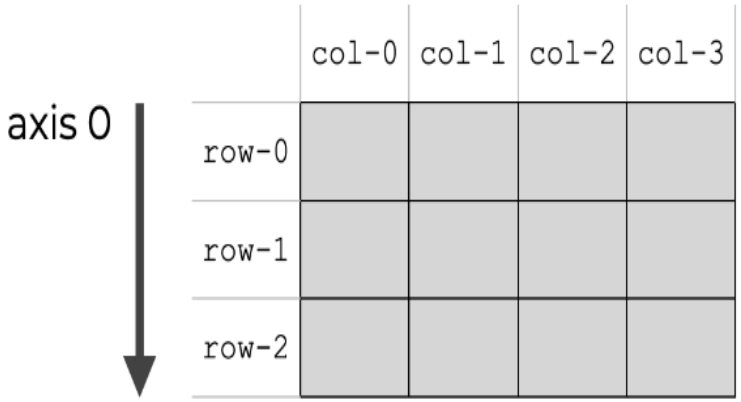
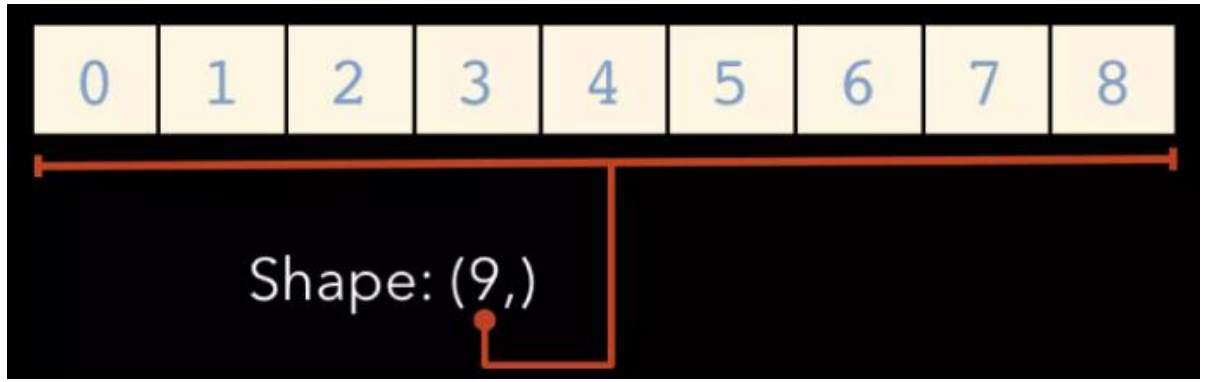
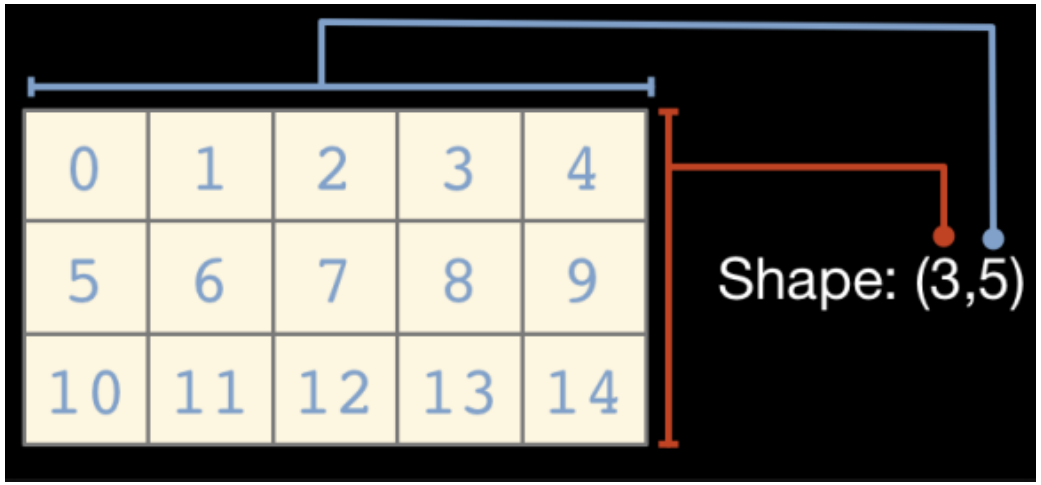
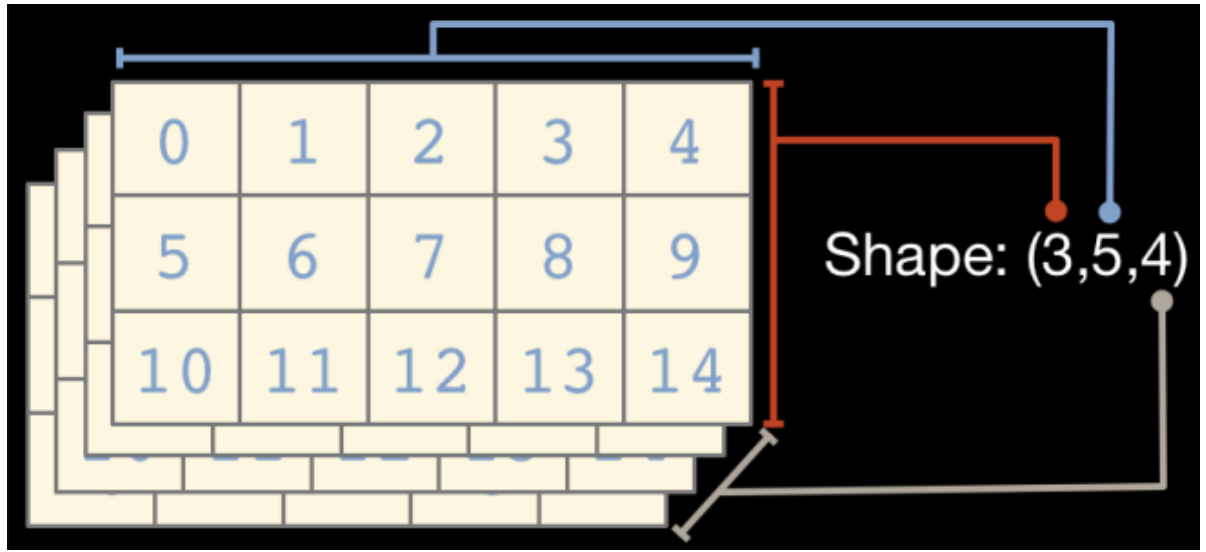


Axis in 2D NumPy Array

When we're talking about 2-d and multi-dimensional arrays, axis 1 is the axis that runs horizontally across the columns.



Assuming that we're talking about multi-dimensional arrays, axis 0 is the axis that runs downward down the rows.



Arrays in NumPy